

SQL and Tableau for Data Visualization - Mini Project Guidelines

Course: SQL and Tableau for Data Visualization

Total Marks: 30 (Approx. 15–18 hours effort per group)

Submission: Tableau Workbook (.twb/.twbx) + SQL Script (.sql) + Final Report (PPT)

1. Project Objective

The objective of this project is to apply SQL querying techniques and Tableau visualization skills for real-world business data analysis. Students will work with publicly available datasets to perform:

- Data extraction, transformation, and aggregation using SQL
- Advanced SQL querying using GROUP BY, HAVING, JOINS, Subqueries, and Window Functions
- Interactive dashboard creation using Tableau
- KPI analysis and statistical visualization
- Trend analysis and business insight generation

Students must demonstrate the ability to integrate SQL data processing workflows with Tableau visual analytics techniques to derive meaningful insights from datasets.

2. Project Workflow

2.1 Dataset Collection and Understanding

- Download datasets from Kaggle, UCI Repository, World Bank, or other public sources
- Understand dataset attributes and business context
- Identify dimensions and measures
- Describe dataset size, structure, and key variables

2.2 SQL Data Preparation and Analysis

- Perform SELECT queries for data retrieval
- Apply WHERE filtering conditions
- Use GROUP BY and HAVING clauses for aggregations
- Implement JOIN operations across multiple tables
- Write Subqueries for advanced filtering and analysis
- Use Window Functions such as RANK(), DENSE_RANK(), ROW_NUMBER(), and RUNNING_TOTAL

2.3 Tableau Visualization and Dashboard Development

- Create row-level and table calculations
- Apply aggregation functions like SUM(), AVG(), COUNT()

- Use Level of Detail (LOD) expressions
- Build line charts with trend lines and forecasting
- Design KPI dashboards and dual-axis charts
- Create bubble charts and geographic maps
- Implement cascading filters and custom actions
- Develop interactive storyboards for business storytelling

2.4 Statistical and Business Analysis

- Perform Pareto analysis for identifying key contributors
- Analyze customer segments using RFM analysis
- Study sales trends, profitability, and regional performance
- Generate business insights using visualization outputs

2.5 Insight Generation

- Identify key trends and patterns
- Provide business recommendations
- Interpret dashboard findings
- Present data-driven conclusions

3. Recommended Tableau Visualizations

- Line Charts with Forecasting
- Dual-Axis Charts
- Bubble Charts
- Pareto Charts
- Maps and Geographic Analysis
- KPI Dashboards
- Storyboards
- Custom Interactive Dashboards

4. Evaluation Rubrics

Component	Description	Marks
Dataset Understanding	Dataset explanation, variables, and business context	3
SQL Querying	Usage of JOINS, GROUP BY, HAVING, Subqueries, Window Functions	5
Tableau Dashboard Design	Visualization quality, dashboard interactivity, storytelling	6
Statistical / KPI Analysis	Trend analysis, Pareto, RFM, KPI interpretation	4
Insight Generation	Business findings and recommendations	4

Documentation and Report	Structure, formatting, and explanation quality	4
Presentation and Viva	Presentation clarity and technical understanding	4

5. Project Presentation (PPT) Guidelines

- Slide 1: Project Title and Team Details
- Slide 2: Problem Statement and Objective
- Slide 3: Dataset Description
- Slide 4: SQL Queries and Data Preparation
- Slide 5: Tableau Dashboard and Visualizations
- Slide 6: Key Findings and Insights
- Slide 7: Conclusion and Recommendations

6. Recommended Project Topics

- Retail Sales and Profitability Dashboard
- Customer Segmentation using RFM Analysis
- Regional Sales Performance Analysis
- Discount Impact on Revenue and Profit
- E-commerce Customer Behavior Analysis
- Inventory and Supply Chain Performance Dashboard
- Banking KPI and Loan Risk Analysis
- Healthcare Data Trend Analysis
- Financial Performance and KPI Dashboard
- Employee Performance and Attrition Analysis

7. Expected Deliverables

- SQL Script File (.sql)
- Tableau Workbook (.twb/.twbx)
- Final PPT Presentation
- Project Report Documentation